Or alternatively,

$$MI = b_1 \cdot I^*_{\$} \cdot \frac{Q}{Q+V} + (1-b_1) \cdot I^*_{\$} \tag{4.10}$$

We can see from Equation 4.10 that if participant A transacts more aggressively, say in a shorter period where only half of the expected market volume will transact, that is, $\frac{1}{2}V$, the temporary market impact cost allocated to participant A will now be:

$$\frac{Q}{Q+\frac{1}{2}V}$$

which is a higher percentage than previously.

If A trades over a longer period of time where $2V$ shares are expected to trade, market impact cost allocated to them will be:

$$\frac{Q}{Q+2V}$$

which is a smaller percentage than previously.

This example helps illustrate that market impact cost is directly related to the urgency of the strategy. Quicker trading will incur higher costs on average than slower trading which will incur lower costs on average. Trading risk, on the other hand, will be lower for the more urgent orders and higher for the more passive orders, i.e. the trader's dilemma.

Due to the rapidly changing nature of the financial markets from regulatory change, structural changes, and investor confidence and perception of order flow information that often accompanies aggressive trading, many participants have begun to fit the market impact model using a more general form of the equation that incorporates an additional parameter a_4. This formulation is:

$$MI_{bp} = b_1 \cdot I^* \cdot \left(\frac{Q}{Q+V}\right)^{a_4} + (1-b_1) \cdot I^* \tag{4.11}$$

Or in terms of POV we have:

$$MI_{bp} = b_1 \cdot I^* \cdot POV^{a_4} + (1-b_1) \cdot I^* \tag{4.12}$$

The relationship between temporary impact and POV rate is shown in Figure 4.5. The percentage of temporary impact that will be allocated to the order is shown on the y-axis and the corresponding POV rate is shown on the x-axis. The figure shows the percentage allocated for various POV functions For example, when $a_4 = 1$ the relationship is linear and costs